

Introduction to Statistics

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Outline

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Introduction

What is Statistics?

- Collecting data
- Gaining insights from data
- Making decisions based on the insights gained from the data

Definition

Statistics can be defined as the art and science of:

- collecting, cleaning and organizing data
- summarizing and analyzing data
- presenting the summary or the analysis
- interpreting the analysis results
- gaining insights through analysis of data
- and finally, drawing valid conclusions and making sound decisions through the use of data.

Scope of Statistics with Examples

- Medicine: evaluating the effectiveness of a new drug using clinical trial data
- Epidemiology: assessing the rate at which a disease spreads
- Education: analyzing exam results to compare teaching methods
- Economics: studying inflation unemployment and income trends
- Business: forecasting sales and analyzing customer behavior
- Engineering: monitoring process quality and detecting defects
- Social sciences: understanding public opinion through surveys

Statistics in Engineering

- Monitoring and controlling manufacturing process quality using control charts
- Analyzing failure rates and predicting the reliability and lifespan of components
- Designing experiments to optimize product performance and material properties
- Detecting defects and reducing variability in production processes
- Modeling and simulating systems to evaluate engineering designs

Statistics in Biotechnology

- Analyzing gene expression data
- Comparing growth rates of genetically modified and non modified organisms
- Estimating mutation rates from DNA sequencing data
- Assessing the effectiveness of gene therapy treatments
- Designing and conducting experiments for drug and vaccine development

Some Basic Statistical Concepts

Population and Sample

- **Population** is the collection/set of all items or individuals of interest in a given study
- **Sample** is a *representative* portion of the population

For example:

- A study may target all the people in Bangladesh. However, it is unfeasible to collect information of everyone in the country in a timely or cost-effective way
- Therefore, data is collected from only a small portion of people from *all over the country*, this is called sampling. The individuals in a sample are usually selected randomly

Census and Survey

- Census:
 - Data are collected from every unit in the population
 - Provides complete information with no sampling error
 - Usually time consuming and expensive
 - Suitable for small or well defined populations
- Survey:
 - Data are collected from a sample of the population
 - Results are subject to sampling variability
 - Faster and more cost effective
 - Suitable for large or infinite populations
 - Allows estimation and inference about the population

Parameter and Statistic

- A **parameter** is a characteristic or function of every objects or individuals in a population. For a fixed population, it is a fixed (but, usually unknown) value
- A **statistic** is a characteristic or function of every objects or individuals in a sample.
- A **statistic** is used to *estimate* a **parameter**

Parameter and Statistic (cont.)

- For a fixed population, the value of a parameter is fixed (but usually unknown)
- However, due to randomization, different samples can include different individuals from a population
- Therefore, the value of a statistic can vary across different samples

Parameter and Statistic (cont.)

For example:

- Suppose the goal is to find the average height of the students of a class
- The population average is a fixed value and it is unknown unless data is collected from everyone in the class
- If the heights of a some students are collected as a random sample, then we can estimate the population average using the sample average
- If another sample is collected, the same individuals as the first sample may not be selected, therefore, the estimate shall be different from the first estimate

Types of Statistics

- **Descriptive statistics:** Methods for organizing, summarizing and presenting data in an informative way. For example:
 - A hypothetical customer survey finds that 50% of the customers are satisfied with a product
- **Inferential statistic:** Methods for using sample data to make predictions, test hypotheses, and generalize conclusions about a larger population. For example:
 - A study finds association between smoking and cancer

Variable and Measurement

Variable

- Variable means something that can vary
- It is a characteristic that can vary across individuals or objects or items or cases of a phenomenon
- For example:
 - Age
 - Gender
 - Socioeconomic status
 - Temperature

Types of Variables

Variables:

- Qualitative
- Quantitative:
 - Discreet
 - Continuous

Qualitative vs Quantitative

- **Qualitative:** Qualitative variables describe qualities and are categorical. These are non-numerical and descriptive values that represent attributes or categories. For example:
 - Name of a person
 - Gender
 - Hair colour
- **Quantitative:** Quantitative variables measure quantities with numbers. These are numeric data that can be counted or measured, allowing for mathematical calculations. For example:
 - Height
 - Temperature
 - Number of students in a class

Discreet vs Continuous

- **Discreet:** *Countable*, usually whole numbers. Finite number of possible values. For example:
 - Number of students in a class
- **Continuous:** *Measurable*, can have fractional values. Can take values in a given range. Infinite number of possible values in any range. For example:
 - Height

Scales of Variables or Level of Measurement

Scales of variables, or levels of measurement, define how data is categorized and quantified/measured. It dictates which statistical analysis is appropriate for a variable.

There are four levels of measurement:

- Nominal
- Ordinal
- Interval
- Ratio

Nominal and ordinal are for categorical or qualitative data, and the last two are for quantitative data.

Nominal

- Nominal scale is for variables with categories with no inherent order or ranking (labels/names only)
- Example:
 - Gender
 - Religion
- Allowed operations:
 - We can only **count** the occurrence or frequency of each of the categories of a nominal variable

Ordinal

- Ordinal scale is for variables with categories that can be *ordered* or ranked, but differences between ranks are not necessarily equal or measurable
- Example:
 - Satisfaction level (low, medium, high)
 - Race position (first, second, third)
 - Socioeconomic status (lower, mid, upper)
- Allowed operations:
 - We can **count** the frequency of each of the categories of an ordinal variable as well as **order** the categories

Interval

- It is for numerical variables. The values are ordered, with equal, meaningful *intervals* between values
- There is no true zero point
- Example:
 - Temperature (0°C does not imply the lack of heat or thermal energy)
 - Years on a Calendar
- Since there is no true zero point, taking ratio of two values of an interval scale variable is meaningless
- Allowed operations:
 - We can **add**, **subtract** different values of an interval scale variable

Example: Interval Scale Variables

- **Temperature (°C or °F):** The difference between 20°C and 30°C is the same as between 30°C and 40°C — intervals are equal and meaningful. However, 0°C does not mean “no temperature,” so ratios are meaningless: 40°C is *not* twice as hot as 20°C
- **IQ Score:** The difference between scores of 90 and 100 is the same as between 110 and 120. However, a score of 0 does not mean the complete absence of intelligence, so we cannot say someone with IQ 140 is “twice as intelligent” as someone with IQ 70
- **Calendar Year:** The gap between year 1900 and 1950 equals that between 1950 and 2000. However, year 0 is an arbitrary reference point, not a true absence of time, so saying year 2000 is “twice as recent” as year 1000 is meaningless

Ratio

- An ratio scale variable includes all properties of interval scales, plus a true zero point, allowing for meaningful ratios
- Example:
 - Height
 - Weight
 - Income
- Allowed operations:
 - All arithmetic operations (addition, subtraction, multiplication, division, ratios)

Example: Ratio Scale Variables

- **Height:** A height of 0 cm means the complete absence of height. Equal intervals exist between measurements, and ratios are meaningful — a person who is 180 cm is exactly twice as tall as a child who is 90 cm
- **Weight:** Zero weight means no mass at all. A package weighing 10 kg is genuinely twice as heavy as one weighing 5 kg, making both differences and ratios interpretable
- **Income:** Zero income means no earnings whatsoever. Someone earning \$60,000 per year earns exactly three times as much as someone earning \$20,000, which is a meaningful and interpretable ratio
- **Reaction Time:** A reaction time of 0 would indicate an instantaneous response (true zero). A response time of 400 ms is twice as long as one of 200 ms, and this comparison is physically meaningful

Sources of Data

Data

- Data in statistics consists of numerical or qualitative facts collected for analysis
- They are recorded observations about individuals, objects, or events
- Data arise from measurement, counting, surveys, or experiments
- Each observation consists of one or more variables
- Usually presented in a tabular form, can also be in other forms (such as images, audio, video etc.)
- Statistics uses data to summarize information and draw conclusions about a larger context

Sources of Data

- **Primary data:** data collected firsthand by the researcher for a specific study
- **Secondary data:** data previously collected by others for a different or a more general purpose
- **Tertiary data:** data that summarize, compile, or index primary and secondary sources

Primary Data

Data collected firsthand by the researcher for a specific study.

- Advantages:
 - Directly relevant to the research objectives
 - Greater control over data quality and measurement methods
 - Up-to-date and specific to the population of interest
 - Clear understanding of how the data were collected
- Disadvantages:
 - Time-consuming to collect
 - Often expensive in terms of money and resources
 - Requires careful planning and technical expertise
 - Limited scope compared to large existing datasets

Secondary Data

Data previously collected by others for a different or a more general purpose.

- Advantages:
 - Quick and inexpensive to obtain
 - Often covers large populations or long time periods
 - Useful for comparisons and trend analysis
 - No need for data collection infrastructure
- Disadvantages:
 - May not exactly match the research objectives
 - Limited control over data quality and measurement methods
 - Possible issues with outdated or incomplete data
 - Documentation and variable definitions may be unclear

Other Ways to Classify Data

- Observational data
- Experimental data
- Administrative records/registers
- Publications and reports

Observational Data

Data collected without controlling or manipulating variables.

- Variables are observed as they naturally occur
- Common in surveys and observational studies
- Advantages:
 - Easier and cheaper to collect
 - Suitable when experiments are impractical or unethical
 - Reflects real world conditions
- Disadvantages:
 - Cannot establish cause and effect relationships
 - More vulnerable to confounding factors

Experimental Data

Data generated by actively manipulating one or more variables.

- Presence of control and treatment groups
- Random assignment of experimental units
- Advantages:
 - Allows identification of cause and effect relationships
 - Greater control over experimental conditions
 - Reduces the impact of confounding variables
- Disadvantages:
 - Often time consuming and expensive
 - May face ethical or practical limitations
 - Artificial settings may reduce realism

Questions?
